

PROPERTIES USED TO EXPAND LOG EXPRESSIONS

Product Property

$$\log(xy) = \log x + \log y$$

Quotient Property

$$\log\left(\frac{x}{y}\right) = \log x - \log y$$

Power Property

$$\log x^n = n \log x$$

$$\log(5a) = \log 5 + \log a$$

$$\log\left(\frac{3}{x+1}\right) = \log 3 - \log(x+1)$$

$$\log x^2 = 2 \log x$$

PROPERTIES USED TO CONDENSE LOG EXPRESSIONS

Product Property

$$\log x + \log y = \log(xy)$$

Quotient Property

$$\log x - \log y = \log\left(\frac{x}{y}\right)$$

Power Property

$$n \log x = \log x^n$$

$$\log_4 2 + \log_4 x = \log_4(2x)$$

$$\log 7 - \log 2 = \log\left(\frac{7}{2}\right)$$

$$3 \log_2 x = \log_2 x^3$$

Expand logarithms

$$1. \log(x^5 y^2) = \log x^5 + \log y^2 = 5 \log x + 2 \log y$$

$$\star 3. \log_3 7(2x-3)^2 = \log_3 7 + \log_3 (2x-3)^2 = \log_3 7 + 2 \log_3 (2x-3)$$

Condense logarithms

$$2. 3 \log x - 2 \log y = \log x^3 - \log y^2 = \log\left(\frac{x^3}{y^2}\right)$$

$$\star 4. 2 \log_7 x + \log_7 y - 3 \log_7 z = \log_7 x^2 + \log_7 y - \log_7 z^3 = \log_7 (x^2 y) - \log_7 z^3 = \log_7 \left(\frac{x^2 y}{z^3}\right)$$

Ex. Condense: $k \log c - y \log d$

$$\log c^k - \log d^y = \log\left(\frac{c^k}{d^y}\right)$$

5. Condense: $k \log c + r \log d$

$$\log c^k + \log d^r = \log(c^k \cdot d^r)$$

Expand the logarithm fully using the properties of logs. Express the final answer in terms of $\log x$, $\log y$, and $\log z$.

$$\sqrt[3]{x^2} = x^{\frac{2}{3}} \quad \begin{matrix} \leftarrow \text{power} \\ \leftarrow \text{root} \end{matrix}$$

$$\sqrt{y} = y^{\frac{1}{2}}$$

$$\log \frac{z\sqrt{y}}{x^2} = \log \frac{zy^{\frac{1}{2}}}{x^2}$$

$$\log zy^{\frac{1}{2}} - \log x^2 = \log z + \log y^{\frac{1}{2}} - \log x^2 = \log z + \frac{1}{2} \log y - 2 \log x$$

Expand the logarithm fully using the properties of logs. Express the final answer in terms of $\log x$, $\log y$, and $\log z$.

$$\log \frac{y^5}{z^2 \sqrt{x^3}}$$

$$\begin{aligned} &= \log \frac{y^5}{z^2 \cdot x^{\frac{3}{2}}} \\ &= \log y^5 - \log z^2 \cdot x^{\frac{3}{2}} \\ &= \log y^5 - [\log z^2 + \log x^{\frac{3}{2}}] \\ &= 5 \log y - 2 \log z - \frac{3}{2} \log x \end{aligned}$$

Alg2 Notes 7-4: Properties of Logarithms (notes on backside)

Warm-Up: Find the value of:

1. $\log_4 16 = 2$

2. $\log_7 7 = 1$

3. $\log_3 1 = 0$

4. $\log_2 \frac{1}{8}$

$\log_2 2^{-3} = -3$

5.

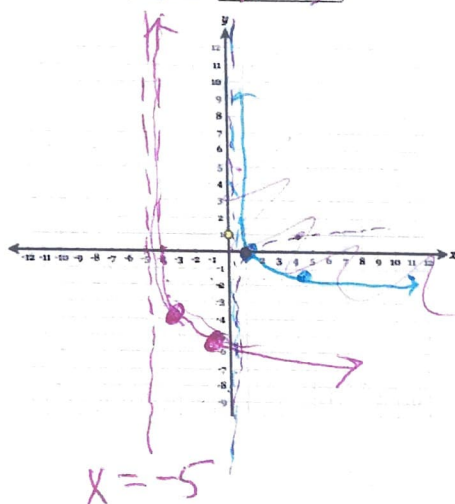
flip over x

left 5

Down 4

Graph the function $f(x) = -\log_4 (x + 5) - 4$ on the axes below. You must plot the asymptote and any two points with whole number coordinates.

Asymptote: $x = -5$



1st $\log_4 x = 2$
 $\log_4 1 = 0$
 $\log_4 4 = 1$

2nd flip it

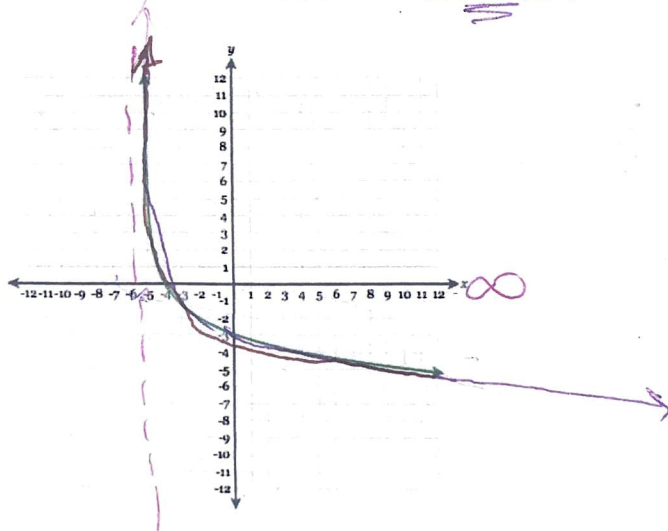
3rd left 5 down 4

$2 \Rightarrow \frac{1}{8}$

$(1, 0)$
 $(4, 1)$

6.

What are the features of the function $f(x) = -2 \log_3 (x + 5)$ graphed below?



The function $f(x)$ is $\log$ function with a vertical asymptote of

$x = -5$. The range of the function is $(-\infty, \infty)$ and it is decreasing on its

domain of $(-5, \infty)$. The end behavior on the LEFT side is as

$x \rightarrow -5$ $y \rightarrow \infty$ and the end behavior on the RIGHT side is as

$x \rightarrow \infty$ $y \rightarrow -\infty$